

$R = ax + by + c$ pro (x_i, y_i, z_i) platí $z_i \approx ax_i + by_i + c$
 koeficienty a, b, c nově náhodná matice A
 $Ax = R$ $ax_i + by_i + c \Rightarrow (x_i, y_i, 1)$

$$u = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$A = \begin{pmatrix} x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots \\ x_m & y_m & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

normální rovnice $A^T A u = A^T R$

$$A^T A = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \quad A^T R = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$$

$$a = \frac{1}{2} \quad b = \frac{3}{4} \quad c = \frac{1}{2}$$

$$\boxed{R = \frac{1}{2}x - \frac{3}{4}y + \frac{1}{2}}$$